GRADE 12 / Mathematics / English

Differentiate and interpret

01 Understand

Learning goals: Differentiate polynomial terms • Find stationary points • Use a sign test to classify turning points

The derivative $f'(x)$ describes the gradient of the tangent. For a power ax^n, the derivative is anx^{n-1}.

At a stationary point $f'(x) = 0$. Use $f(x)$, not $f'(x)$, to find the point's y-coordinate.

A change in f' from positive to negative indicates a local maximum; negative to positive indicates a local minimum.

Gradient sign chart for $f(x) = x^3 - 3x$

Interval	Sign of f'	Behaviour
$x < -1$	+	increasing
$-1 < x < 1$	−	decreasing
$x > 1$	+	increasing

02 Follow the reasoning

Find and classify the stationary points of $f(x) = x^3 - 3x$.

1. $f'(x) = 3x^2 - 3$. Set $3(x^2 - 1) = 0$, so $x = -1$ or 1.

2. $f(-1) = 2$ and $f(1) = -2$. The points are $(-1; 2)$ and $(1; -2)$.

3. The derivative changes + to − at -1: local maximum. It changes − to + at 1: local minimum.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Differentiate $2x^3 - 5x^2 + 4$.

2. Find the gradient of its tangent at $x = 2$.

3. Find the tangent equation there.

04 Check and correct

1. $6x^2 - 10x$. The derivative of the constant is zero.

2. $6(2)^2 - 10(2) = 4$.

3. $f(2) = 0$, so $y - 0 = 4(x - 2)$, giving $y = 4x - 8$.

Common mistake to avoid

$f'(a)$ is a gradient, not the height of the graph. Find $f(a)$ separately.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Differential calculus: derivatives; cubic functions

DBE CAPS, Grades 10–12: Mathematics. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID math-12.